

Urban Night Mobility Data Warehouse

NYC Citi Bike night mobility analysis with weather, holidays, and geographic enrichment

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- 📄 Full reproducible data warehousing pipeline built with PostgreSQL/PostGIS and Python.



Project Goal & Research Questions

Build a **reproducible data warehouse** to analyze night-time Citi Bike mobility in New York City during 2024, integrating weather, holidays, and geographic boundaries. **January 2024** is the validated development sample; the pipeline is designed for full-year execution.

Spatial Concentration

Where is night mobility concentrated across NYC boroughs?

Temporal Variation

How does night demand vary by weekday, weekend, and holiday windows?

Weather Impact

How do weather conditions affect night trip volumes?

Borough Flows

Which origin–destination borough flows are most relevant at night?

Four Heterogeneous Data Sources

Source	Content	Purpose
Citi Bike trip data	Trip-level bike-sharing records (start/end stations, timestamps, rider type)	Primary mobility fact source
NOAA daily weather (Central Park)	Temperature, precipitation, snow, and weather attributes	Enrich trips by environmental conditions
US public holiday calendar	US public holidays for 2024	Identify holiday and near-holiday mobility windows
NYC borough / NTA boundaries	Spatial polygons for boroughs and neighborhoods	Assign trips and stations to geographic zones via PostGIS

 Raw files are stored in `data_raw/` and ignored by Git to ensure reproducibility and manage storage.

End-to-End Pipeline Architecture



Four PostgreSQL schemas enforce clear layer separation: **staging** (raw source loads), **reconciled** (cleaned and integrated relational data), **dw** (dimensional facts and dimensions), and **audit** (row counts, load metadata, and quality results).

ETL Operations & Reconciled Layer

Raw Sources

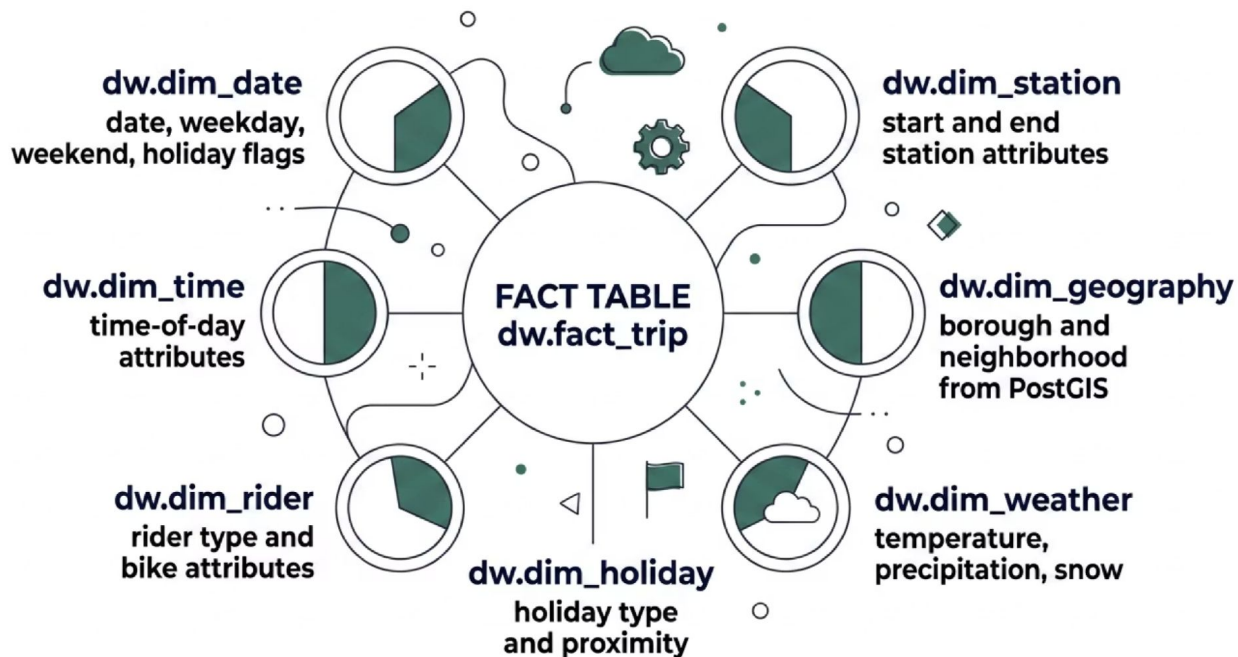
- CSV, JSON, and geospatial files
- Inconsistent timestamps and identifiers
- Incomplete or invalid trip records
- No geographic or holiday enrichment

Reconciled Layer

- Standardized timestamps and station identifiers
- Invalid or incomplete records cleaned or rejected
- Stations and trips integrated with borough geography
- Date attributes enriched: weekday, weekend, holiday flags
- Consistent relational tables ready for warehouse loading

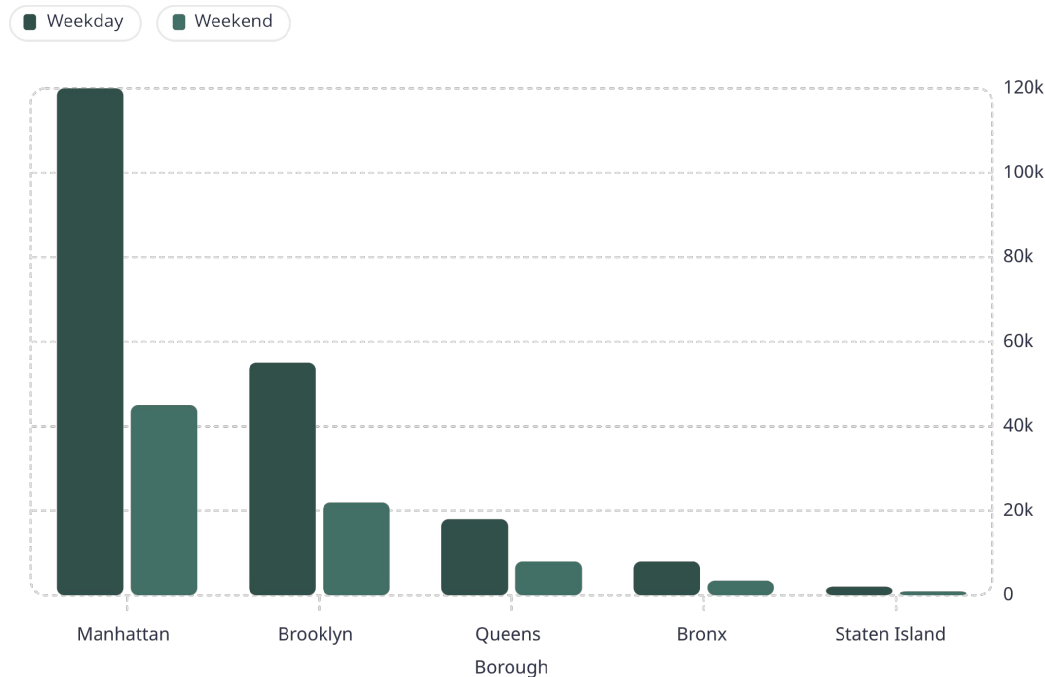
Dimensional Model & Physical Schema

The central analytical event is a **night bike trip**, with measures including trip count, duration, and aggregated demand. The **star schema** supports OLAP operations: **slice**, **dice**, **roll-up**, and **drill-down**.



The schema is optimized for analytical queries — not raw transaction storage. All dimensions use surrogate keys; the fact table references them for efficient OLAP aggregation.

OLAP Analysis: Night Demand by Borough & Day Type



OLAP Operations

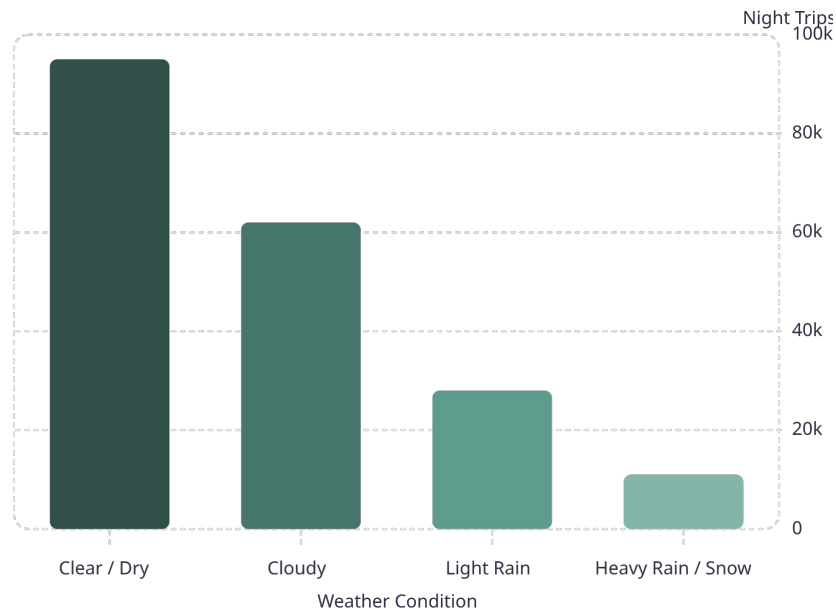
- **Slice:** filter to night trips only
- **Roll-up:** aggregate demand to borough level
- **Dice:** compare weekdays vs weekends vs holidays

Key Finding

Night mobility is **strongly concentrated in Manhattan**, followed by Brooklyn. Weekday demand consistently exceeds weekend demand across all boroughs in the January 2024 sample.

Weather Impact & Borough Origin–Destination Flows

Weather & Night Trip Demand



Slice + aggregation + comparison: adverse weather significantly reduces night trip volumes. Weather enrichment connects mobility behavior with environmental context using the same fact table.

Top Borough OD Flows (January 2024)

Start Borough	End Borough	Night Trips
Manhattan	Manhattan	~95,000
Brooklyn	Brooklyn	~38,000
Manhattan	Brooklyn	~22,000
Brooklyn	Manhattan	~20,000
Queens	Manhattan	~12,000

Roll-up + ranking: most night trips remain intra-borough. Inter-borough flows reveal key mobility corridors between Manhattan and Brooklyn.